MUHAMMAD HASSAN ADIL

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SUMMARY

As a Computer Science undergraduate passionate about AI and Machine Learning, I bring strong skills in Python, data analysis, and model deployment using tools like TensorFlow and Scikit-learn. I enjoy blending analytical thinking with creative pursuits like video editing and production. Winner of the Best Final Year Project Award for an ML-powered IoT fall detection system, I aim to contribute to innovative, data-driven teams creating social impact.

EDUCATION

BSc Computer Sciences Graduated Jan 2025
Forman Christian College (A Chartered university) (FCCU), Punjab, PK
Beacon-house College Campus Gulberg -*A Levels*

SKILLS

Technical Proficiencies: Java, SpringBoot, Python, MySQL, TensorFlow, Git/Github, AWS, Jupyter Notebook
Creative and Design Skills: Adobe Photoshop , Adobe Premiere Pro , Adobe Lightroom , Canva

EXPERIENCE

Hotelkey, Lahore, PK: Software Engineer July 2025 - Present

– Led software development for Central Reservation System (CRS) by debugging live production issues, implementing booking and inventory management logic, collaborating with QA and product teams to ensure seamless integration with major OTAs like **Airbnb**, **Expedia**, and **Booking.com**.

Forman Christian College (A Chartered University), Lahore, PK: Lab Engineer Feb - June (2025)

– Created and conducted labs for Java, Python and C++. Helped students understand and implement important programming concepts including Object Oriented Programming and design of algorithms.

Hotelkey, Lahore, PK: Part-Time Software Engineer March - July (2024)

– Developed data-driven integration tests in JUnit for travel platforms like Airbnb and Expedia, analyzed backend logs to ensure accurate data synchronization with AWS-hosted systems.

Hotelkey, Lahore, PK: Software Engineer Intern July – August (2023)

– Engineered backend modules using Java SpringBoot and developed automated API tests with REST Assured and Postman for internal productivity tools.

ACADEMIC PROJECTS

Fall Protection Vest (Best Final Year Project Award) Fall 2024
Built an ML-powered wearable vest that detects falls using sensor fusion and triggers mobile alerts.

– Trained classification models on the UMAFall dataset using Scikit-learn, integrated with ESP32 and MPU9250 sensors, and connected to a Flutter app for real-time fall alerts.

Fetal Health Classification Fall 2022
Classified fetal health conditions using machine learning techniques on medical data.

– Used Pandas and Scikit-learn to clean data and apply KNN, evaluating model accuracy with confusion matrices and visualizations in Matplotlib.

Search Engine Development Spring 2022
Built a custom search engine in Python to compare data structure performance.

– Implemented indexing and retrieval logic with hash tables, linked lists, and queues, followed by benchmarking for performance insights.

Healthcare Management System Fall 2021
Designed a GUI-based Python system for healthcare administration.

– Developed an OOP-based Tkinter app for managing appointments, billing, and patient records with integrated validation and search functionality.

Facial Recognition System Spring 2021
Built a facial recognition pipeline using Python and OpenCV for identity verification.

– Preprocessed a custom dataset and implemented feature-based classification using Scikit-learn in Google Colab.